

SHETH L.U.J AND SIR M.V COLLEGE

PRACTICAL NO :- 11

AIM :- Reshaping data using pivot_longer()/pivot_wider() (R).

INPUT :-

```
library(dplyr)
```

```
library(tidyr)
```

```
df <- spotify_data.clean %>%
```

```
  mutate(ProductID = row_number()) %>%
```

```
  select(ProductID, artist_name, track_popularity, artist_popularity)
```

```
print("--- 1. Original Wide Data ---")
```

```
print(head(df))
```

```
long_df <- df %>%
```

```
  pivot_longer(
```

```
    cols = c(track_popularity, artist_popularity),
```

```
    names_to = "Metric",
```

```
    values_to = "Value"
```

```
)
```

```
print("--- 2. Long Format (pivot_longer) ---")
```

```
print(head(long_df, 6))
```

```
wide_df <- long_df %>%
```

```
  pivot_wider(
```

```
    names_from = Metric,
```

```
    values_from = Value
```

```
)
```

```
print("--- 3. Wide Format (Back to Original) ---")
```

```
print(head(wide_df))
```

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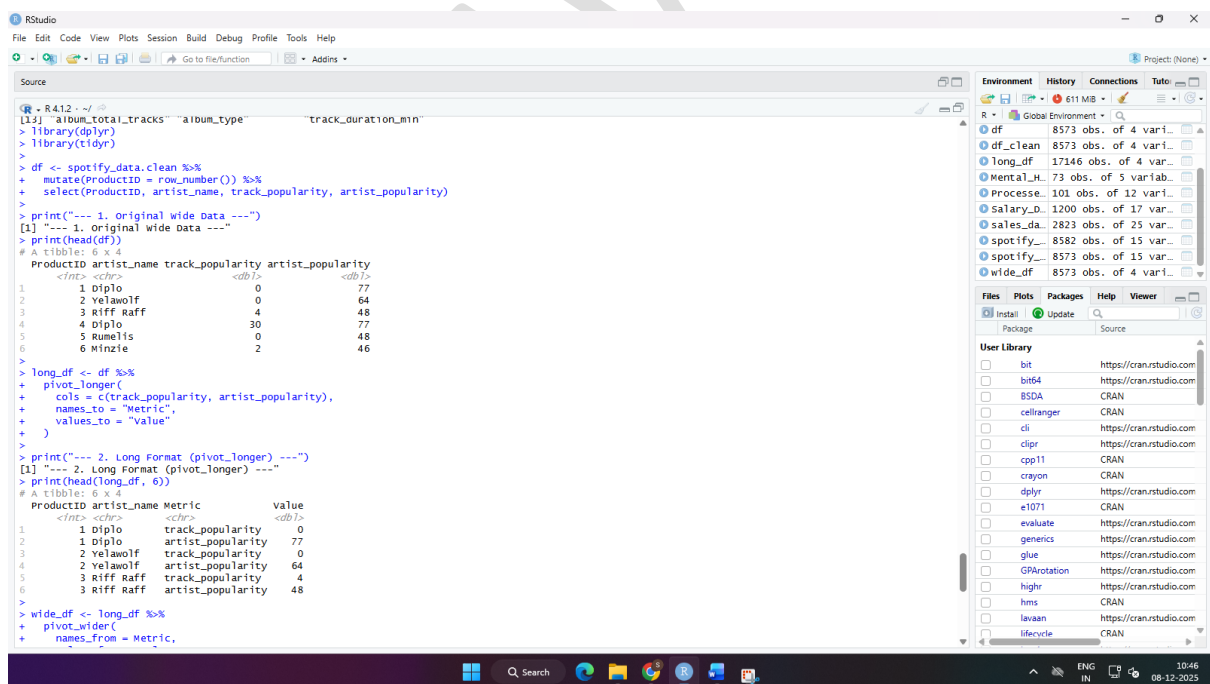
ROLL NO :- S084

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```
df_clean <- df %>%  
  
mutate(artist_name = ifelse(is.na(artist_name), "Unknown", artist_name))  
  
category_pivot <- df_clean %>%  
  
select(ProductID, artist_name, track_popularity) %>%  
  
pivot_wider(  
  
  names_from = artist_name,  
  
  values_from = track_popularity  
  
)  
  
print("--- 4. Category Pivot (Spreading Categories) ---")  
print(head(category_pivot))
```

OUTPUT :-



The screenshot shows the RStudio interface. The Source pane on the left contains the following R code:

```
R412 -> df <- spotify_data_clean %>%  
+ mutate(ProductID = row_number()) %>%  
+ select(ProductID, artist_name, track_popularity, artist_popularity)  
+  
+ print("--- 1. Original Wide Data ---")  
+ print(head(df))  
# A tibble: 6 x 4  
  ProductID artist_name track_popularity artist_popularity  
  <int> <chr> <dbl> <dbl>  
1 1 Diplo 0 77  
2 2 Yelawolf 64 64  
3 3 Riff Raff 4 48  
4 4 Diplo 30 77  
5 5 Rumelis 0 48  
6 6 Minzle 2 46  
+  
+ long_df <- df %>%  
+ pivot_longer(  
+   cols = c(track_popularity, artist_popularity),  
+   names_to = "Metric",  
+   values_to = "Value"  
+ )  
+  
+ print("--- 2. Long Format (pivot_longer) ---")  
+ print(head(long_df, 6))  
# A tibble: 6 x 4  
  ProductID artist_name Metric Value  
  <int> <chr> <chr> <dbl>  
1 1 Diplo track_popularity 0  
2 1 Diplo artist_popularity 77  
3 2 Yelawolf track_popularity 64  
4 2 Yelawolf artist_popularity 64  
5 3 Riff Raff track_popularity 4  
6 3 Riff Raff artist_popularity 48  
+  
+ wide_df <- long_df %>%  
+ pivot_wider(  
+   names_from = Metric,
```

The Environment pane on the right shows the following objects:

- df: 8573 obs. of 4 vari...
- df_clean: 8573 obs. of 4 vari...
- long_df: 17146 obs. of 4 var...
- Mental_H: 73 obs. of 5 variab...
- Processe: 101 obs. of 12 vari...
- Salary_D: 1200 obs. of 17 var...
- sales_da: 2823 obs. of 25 var...
- spotify_: 8582 obs. of 15 var...
- spotify_: 8573 obs. of 15 var...
- wide_df: 8573 obs. of 4 vari...

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